

Malachi Mabie

AI Systems Engineer

Carpinteria, CA
✉ malachi@outlook.com
🌐 malachimabie.com
in mlmabie

*Designing simple affordances within complex environments.
Research, training, inference, deployment.*

Professional Experience

- Dec 2024– Present **AI/ML Systems Consultant, Mabie Industries, LLC, Carpinteria, CA**
- Led applied AI/ML consulting across cognitive security, product analytics, and AI-native workflows, including ML and product cognitive-security strategy for Siethar Group and AI ops / analytics for Hammies product marketing
 - Developed public, non-proprietary DXF-to-CAD research for construction and architecture workflows: graph-structured representations, typed constraints, persistent object state, learned repair/search loops, and hard validators
 - Through Mabie Industries, expanded an initial software-architecture consulting scope into software architecture, implementation, debugging, deployment, and product ownership for Drone Institute: 6 production systems solo, thermal CV pipeline, multi-tenant marketplace, webhook-driven operations, and \$35/month infrastructure supporting \$811k ARR potential
 - Built agent/runtime infrastructure with MCP-based tool orchestration, 3-store memory, dynamic subagent spawning, and evaluation loops over a 3,165-note, 11k-edge knowledge system
- Apr 2024– Mar 2025 **Senior AI Research Engineer, RAM Laboratories, Inc., San Diego, CA**
- Principal Investigator, T-UEBA (\$250k SBIR): led end-to-end training and eval on 4×T4 GPUs with FSDP, custom flash attention, continued pretraining, rejection sampling, curriculum training, and RL-shaped evaluation
 - Designed CC-GAT with Hopfield conditioning and conformal prediction; delivered 95% accuracy, 0.887 F1, 15.2ms/graph, 15MB model size, and <3% FPR
 - COG-SAFE (DARPA FACT): built an LLM harness for cognitive security and accountable agent transactions; collaborated with Georgia Tech and represented RAM Labs at Google
 - Turned rapidly changing literature into architectures, technical narratives, and executable plans under severe compute, data, and mission constraints
- Apr 2022– Apr 2024 **AI/ML Research Engineer, RAM Laboratories, Inc., San Diego, CA**
- DARPA AIxCC: vulnerability harness with RoBERTa, GraphCodeBERT, CodeT5++, and LLaMA QLoRA; static-analysis → decompilation → runtime-verification pipeline
 - DAICON: distributed inference for heterogeneous edge devices using Fused Tile Partitioning, custom PyTorch hooks, work-stealing scheduling, and NDN-based shard routing
 - DL-Patcher patent (U.S. 18/375,839); authored 15+ SBIRs including LEAR, HopModKG, Warfighter AI, and VAAST
 - Served as a cross-program technical contributor across cybersecurity, distributed systems, CV, and NLP while mentoring junior researchers on model and pipeline design
- 2013–2022 **Earlier Engineering & Teaching, Multiple Organizations, CA**
- **Applied Medical:** robotic tooling, automation, and process optimization for medical-device manufacturing; \$100k/month savings; intern at 16 → technician → engineer
 - **Inspirit AI:** advanced ML instruction and curriculum in RL, transformers, GANs, and project mentorship
 - **Endura / RoadReader / UCSD Design Lab:** quantized audio CNNs, CV for autonomous sensing, Hyundai robotaxi and Ford mobility research

Education

- 2021 **B.S. Cognitive Science: ML & Neural Computation, UC San Diego**
Reinforcement Learning, Neural Signal Processing, Convex Optimization, Systems Programming, HPC

Technical Expertise

Training

FSDP, Flash Attention
Continued Pretraining
LoRA, QLoRA, Distillation
RL: PPO, DPO, GRPO, DQN, MCTS

Inference & Agents

Distributed Inference
Model Sharding, ONNX
MCP, Memory Systems, Verifiers
Context Engineering, Evals

Stack

Python, C/C++, TypeScript
PyTorch, PyG, Transformers
AWS serverless, Docker
Terraform, FastAPI, Linux

Selected Writing

Social Stereotypes as Distributed Cognitive Systems (2020)—distributed cognition and emergent social structure. *Compression Ecology Between Motivation and Inspiration* (2019)—predictive attention and sensemaking.

Selected Programs

T-UEBA: constrained GNN behavior analytics. **DXF-to-CAD:** graph/geometry ML for construction workflows. **COG-SAFE:** accountable LLM security. **AIxCC:** multi-model vulnerability harness.